Statistical Foundations of Learning

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1 Statistical learning problem and Bayes risk

1.1 Learning problem and risk

Setup. Inputs live in a **feature space** $\mathcal{X}$, outputs in a **label set** $\mathcal{Y}$ (binary classification: $\mathcal{Y} = \{0, 1\}$ or $\{\pm 1\}$; regression: $\mathcal{Y} \subseteq \mathbb{R}$). A **predictor** is a map $h : \mathcal{X} \to \mathcal{Y}$. Given a **training sample** $S = ((x_i, y_i))_{i=1}^m \in (\mathcal{X} \times \mathcal{Y})^m$, a **learner** $\mathcal{A}$ returns a predictor $\hat{h}_S = \mathcal{A}(S)$; it must accept samples of every size $m \geq 1$ (think of a neural network trained by gradient descent on S).

Data model. Learning only makes sense if test data resemble training data. We assume a fixed but unknown distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$: training and test pairs are all drawn from $\mathcal{D}$, independently, so $S \sim \mathcal{D}^m$. Equivalently, $\mathcal{D}$ is specified by the marginal $\mathcal{D}_\mathcal{X}$ of x and the conditional law $\mathbb{P}(y \mid x)$ of the label. A deterministic labelling $y = f(x)$ is the special case of a degenerate conditional; it cannot express that two equal inputs may carry different labels.

Loss and risk. A **loss** $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, \infty)$ penalises a prediction $h(x)$ when the truth is y. The two standard choices are the **0-1 loss** $\mathbb{1}\{h(x) \neq y\}$ for classification and the **squared loss** $(h(x) - y)^2$ for regression.

Definition 1.1 (Risk)**.** The **risk** (generalisation error) of h is its average loss on a new pair from $\mathcal{D}$,

$$L_\mathcal{D}(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}\big[\ell(h(x), y)\big].$$

For a learner, the quantity of interest is $L_\mathcal{D}(\hat{h}_S)$, the expected test error of the predictor learned from S.

Under the 0-1 loss, $L_\mathcal{D}(h)$ is the probability that h misclassifies, $L_\mathcal{D}(h) = \mathbb{P}_{(x,y)\sim\mathcal{D}}(h(x) \neq y)$.

Example 1.2 (Two Gaussian classes)**.** Take $\mathcal{X} = \mathbb{R}$, $\mathcal{Y} = \{\pm 1\}$, both labels with probability $\frac{1}{2}$, and class-conditional laws $\mathcal{N}(-\mu, \sigma^2)$ for $y = -1$ and $\mathcal{N}(\mu, \sigma^2)$ for $y = +1$, with $\mu > 0$. Then $\mathcal{D}_\mathcal{X} = \frac{1}{2}\mathcal{N}(-\mu, \sigma^2) + \frac{1}{2}\mathcal{N}(\mu, \sigma^2)$, and $\mathbb{P}(y = +1 \mid x)$ increases from 0 to 1, crossing $\frac{1}{2}$ at $x = 0$. A threshold rule $h(x) = \operatorname{sign}(x - t)$ errs on the $+1$ points left of t and on the -1 points right of t (fig. 1); moving t away from 0 trades errors on one class for errors on the other. The optimal threshold is $t = 0$, where $\operatorname{sign}(x)$ always predicts the more likely label (theorem 1.6); in practice it has to be found from S alone.

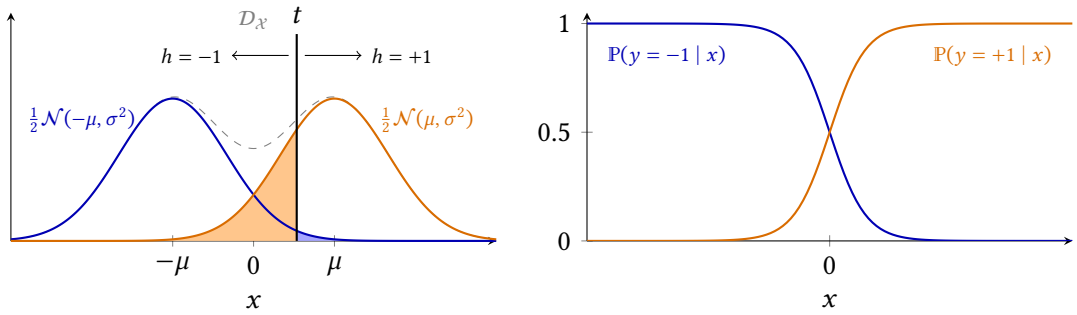


Figure 1: Left: weighted class densities, $\mathcal{D}_\mathcal{X}$ (dashed), errors of $\operatorname{sign}(x - t)$ (shaded). Right: $\mathbb{P}(y = \pm 1 \mid x)$.

1.2 Empirical risk minimisation

Ideally we would solve $\min_{h \in \mathcal{Y}^\mathcal{X}} L_\mathcal{D}(h)$, but $L_\mathcal{D}(h)$ cannot be evaluated without knowing $\mathcal{D}$, so it is replaced by its sample counterpart.

Definition 1.3 (Empirical risk, ERM)**.** The **empirical risk** (training error) of h on S is $L_S(h) = \frac{1}{m}\sum_{i=1}^m \ell(h(x_i), y_i)$. **Empirical risk minimisation** (ERM) solves $\min_{h \in \mathcal{Y}^\mathcal{X}} L_S(h)$.

For fixed h, $L_S(h)$ is an unbiased estimate of $L_D(h)$: each $(x_i, y_i) \sim D$, so by linearity $\mathbb{E}_{S\sim D^m}[L_S(h)] = \frac{1}{m}\sum_{i=1}^m \mathbb{E}\big[\ell(h(x_i), y_i)\big] = L_D(h)$. Minimising it over *all* functions, however, can overfit badly:

Example 1.4 (ERM can fail)**.** Let $x \sim \text{Uniform}([0,1])$ and $y = \mathbb{1}\{x > 0.1\}$. The predictor that outputs 1 on the training points labelled 1 and 0 everywhere else has $L_S(\hat{h}) = 0$. A new x falls in the finite set $\{x_1, \dots, x_m\}$ with probability 0, so $\hat{h}(x) = 0$ almost surely and $\hat{h}$ errs exactly when $y = 1$: $L_D(\hat{h}) = \mathbb{P}(y = 1) = \mathbb{P}(x > 0.1) = 0.9$.

1.3 Bayes risk

Suppose now that D is known. The **Bayes risk** is $L_D^* = \min_{h\in\mathcal{Y}^{\mathcal{X}}} L_D(h)$, and a minimiser h^* is a **Bayes predictor**.

Classification. Let $\mathcal{Y} = \{0,1\}$, ℓ the 0-1 loss, and write $\eta(x) = \mathbb{P}(y = 1 \mid x)$, so that $\mathbb{P}(y = 0 \mid x) = 1 - \eta(x)$.

Theorem 1.5 (Risk of a binary classifier)**.** *For every deterministic $h : \mathcal{X} \to \{0,1\}$,*

$$L_D(h) = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\mathbb{P}(h(x) \neq y \mid x)\big] = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\eta(x)\,\mathbb{1}\{h(x) = 0\} + (1 - \eta(x))\,\mathbb{1}\{h(x) = 1\}\big].$$

Proof. By the tower property,

$$L_D(h) = \mathbb{E}_{(x,y)}\big[\mathbb{1}\{h(x) \neq y\}\big] = \mathbb{E}_x\Big[\mathbb{E}_{y|x}\big[\mathbb{1}\{h(x) \neq y\}\big]\Big] = \mathbb{E}_x\big[\mathbb{P}(h(x) \neq y \mid x)\big].$$

An error occurs either when $h(x) = 0,\ y = 1$ or when $h(x) = 1,\ y = 0$, so $\mathbb{1}\{h(x) \neq y\} = \mathbb{1}\{h(x) = 0\}\mathbb{1}\{y = 1\} + \mathbb{1}\{h(x) = 1\}\mathbb{1}\{y = 0\}$. Given x, $h(x)$ is a constant (h is deterministic), so only the indicators of y are averaged, with $\mathbb{E}_{y|x}[\mathbb{1}\{y = 1\}] = \eta(x)$ and $\mathbb{E}_{y|x}[\mathbb{1}\{y = 0\}] = 1 - \eta(x)$:

$$\mathbb{P}(h(x) \neq y \mid x) = \eta(x)\,\mathbb{1}\{h(x) = 0\} + (1 - \eta(x))\,\mathbb{1}\{h(x) = 1\}. \qquad \square$$

At each x, a classifier therefore pays $\eta(x)$ by predicting 0 or $1 - \eta(x)$ by predicting 1. The best choice is the cheaper option, which defines the **Bayes classifier**

$$h^*(x) = \mathbb{1}\{\eta(x) \geq \tfrac{1}{2}\}.$$

Theorem 1.6 (Bayes risk equals the risk of the Bayes classifier)**.** *For the 0-1 loss,*

$$L_D^* = \mathbb{E}_{x\sim D_{\mathcal{X}}}\big[\min\{\eta(x), 1 - \eta(x)\}\big] = L_D(h^*) = \min_{h\in\{0,1\}^{\mathcal{X}}} L_D(h).$$

Proof. Fix x. Exactly one of $\mathbb{1}\{h(x) = 0\}$, $\mathbb{1}\{h(x) = 1\}$ equals 1, so the integrand of theorem 1.5 is either $\eta(x)$ or $1-\eta(x)$, hence $\geq \min\{\eta(x), 1-\eta(x)\}$. Taking $\mathbb{E}_x$, every h satisfies $L_D(h) \geq \mathbb{E}_x[\min\{\eta(x), 1-\eta(x)\}]$. For h^*: if $\eta(x) \geq \frac{1}{2}$ then $h^*(x) = 1$ and the integrand is $1 - \eta(x) = \min\{\eta(x), 1 - \eta(x)\}$; if $\eta(x) < \frac{1}{2}$ then $h^*(x) = 0$ and it is $\eta(x) = \min\{\eta(x), 1 - \eta(x)\}$. So h^* attains the lower bound, which is therefore the minimum. $\square$

Example 1.7 (Installation crash)**.** Let $x \sim \text{Uniform}[0,1]$ be the completed fraction of a software installation, with $y = 1$ meaning that the system crashes after the fraction x has been completed, and $\eta(x) = |1 - 2x|$. Then $\eta(x) \geq \frac{1}{2}$ iff $x \leq \frac{1}{4}$ or $x \geq \frac{3}{4}$, so $h^* = 1$ on $[0, \frac{1}{4}] \cup [\frac{3}{4}, 1]$ and $h^* = 0$ on $(\frac{1}{4}, \frac{3}{4})$. The function $\min\{\eta, 1 - \eta\}$ is symmetric about $x = \frac{1}{2}$, and on $[0, \frac{1}{2}]$ (where $\eta = 1 - 2x$, $1 - \eta = 2x$) it equals $2x$ for $x \leq \frac{1}{4}$ and $1 - 2x$ for $x \geq \frac{1}{4}$. Hence

$$L_D^* = L_D(h^*) = 2\Big(\int_0^{1/4} 2x\,dx + \int_{1/4}^{1/2} (1 - 2x)\,dx\Big) = 2\Big(\frac{1}{16} + \frac{1}{16}\Big) = \frac{1}{4}.$$

Regression. Let $\mathcal{Y} = \mathbb{R}$ with the squared loss, so that $L_D(h) = \mathbb{E}[(h(x) - y)^2]$. ERM is then least-squares regression (ordinary least squares when h is restricted to linear functions).

Proposition 1.8 (Bayes regressor)**.** *For the squared loss, $h^*(x) = \mathbb{E}[y \mid x]$ and $L_D^* = \mathbb{E}_x\big[\operatorname{Var}(y \mid x)\big]$.*

Proof. Fix x; for deterministic h, $h(x)$ is a constant given x. Expanding the square and then completing it,

$$\begin{aligned}\mathbb{E}\big[(h(x) - y)^2 \mid x\big] &= h(x)^2 - 2h(x)\,\mathbb{E}[y \mid x] + \mathbb{E}[y^2 \mid x] \\ &= \big(h(x) - \mathbb{E}[y \mid x]\big)^2 + \underbrace{\mathbb{E}[y^2 \mid x] - (\mathbb{E}[y \mid x])^2}_{=\operatorname{Var}(y|x)}.\end{aligned}$$

Taking $\mathbb{E}_x$ gives $L_D(h) = \mathbb{E}_x\big[(h(x) - \mathbb{E}[y \mid x])^2\big] + \mathbb{E}_x[\operatorname{Var}(y \mid x)]$. The second term does not depend on h; the first is ≥ 0 and equals 0 for $h(x) = \mathbb{E}[y \mid x]$. □

Example 1.9 (Signal plus noise)**.** If $y = f(x) + \epsilon$ with ϵ independent of x, $\mathbb{E}[\epsilon] = 0$ and $\mathbb{E}[\epsilon^2] = \sigma^2$, then

$$\mathbb{E}[y \mid x] = f(x) + \mathbb{E}[\epsilon] = f(x), \qquad \operatorname{Var}(y \mid x) = \operatorname{Var}(\epsilon) = \mathbb{E}[\epsilon^2] - (\mathbb{E}[\epsilon])^2 = \sigma^2,$$

so by proposition 1.8 $h^* = f$ and $L_D^* = \sigma^2$.

2 Risk of linear regression

2.1 Least squares and the linear model

Ordinary least squares. Let $\mathcal{X} = \mathbb{R}^p$, $\mathcal{Y} = \mathbb{R}$ and $S = ((x_i, y_i))_{i=1}^m \subset \mathbb{R}^p \times \mathbb{R}$. A **linear model** predicts $h_w(x) = x^\top w$ with $w \in \mathbb{R}^p$. The **ordinary least squares** (OLS) estimator is ERM with the squared loss over linear models,

$$\hat{w} = \operatorname*{arg\,min}_{w \in \mathbb{R}^p} L_S(h_w), \qquad L_S(h_w) = \frac{1}{m}\sum_{i=1}^m \big(y_i - x_i^\top w\big)^2.$$

Writing $\hat{\Sigma} = \frac{1}{m}\sum_{i=1}^m x_i x_i^\top$ (a symmetric positive semidefinite $p \times p$ matrix), the minimiser is, whenever $\hat{\Sigma}$ is invertible,

$$\hat{w} = \hat{\Sigma}^{-1}\Big(\frac{1}{m}\sum_{i=1}^m y_i x_i\Big).$$

Ridge regression. When $\hat{\Sigma}$ is singular, the **ridge** estimator with parameter $\lambda > 0$ replaces it by $\hat{\Sigma} + \lambda I$, whose eigenvalues are all $\geq \lambda$, so that it is always invertible. Equivalently, ridge is ERM penalised by the squared Euclidean norm of w:

$$\hat{w}_\lambda = \big(\hat{\Sigma} + \lambda I\big)^{-1}\Big(\frac{1}{m}\sum_{i=1}^m y_i x_i\Big) = \operatorname*{arg\,min}_{w \in \mathbb{R}^p}\ \frac{1}{m}\sum_{i=1}^m \big(y_i - x_i^\top w\big)^2 + \lambda\|w\|^2.$$

Data model. Since y is continuous, its conditional law given x is specified through a model rather than through the probabilities of single values. Let $\mathcal{D}_\mathcal{X}$ be a distribution on $\mathbb{R}^p$ with $\Sigma = \mathbb{E}_x[xx^\top]$, fix $w^* \in \mathbb{R}^p$, and set

$$y = x^\top w^* + \epsilon, \qquad \epsilon \text{ independent of } x, \quad \mathbb{E}[\epsilon] = 0, \quad \mathbb{E}[\epsilon^2] = \sigma^2.$$

Then $\mathbb{E}[y \mid x] = x^\top w^*$ and $\operatorname{Var}(y \mid x) = \sigma^2$, so by proposition 1.8 the Bayes predictor is h_{w^*} and $L_D^* = \sigma^2$. For $S \sim \mathcal{D}^m$ we write $y_i = x_i^\top w^* + \epsilon_i$, where $\epsilon_1, \dots, \epsilon_m$ are i.i.d. copies of ϵ, independent of $x_1, \dots, x_m$. The estimator $\hat{w}$ is a function of S, hence random, and the quantity to compute is the expected risk

$$\mathbb{E}_S\big[L_D(h_{\hat{w}})\big] = \mathbb{E}_S\Big[\mathbb{E}_{(x,y)\sim\mathcal{D}}\big[(y - x^\top \hat{w})^2\big]\Big],$$

with the test pair $(x, y) \sim \mathcal{D}$ independent of S.

2.2 Bias-variance decomposition

Notation. For a positive semidefinite Σ let $\|a\|_\Sigma^2 = a^\top \Sigma a$, and let $\mathrm{trace}(M) = \sum_i M_{ii}$. The covariance matrix of the estimator is

$$\mathrm{Cov}_S(\hat{w}) = \mathbb{E}_S\left[(\hat{w} - \mathbb{E}_S[\hat{w}])(\hat{w} - \mathbb{E}_S[\hat{w}])^\top\right].$$

Theorem 2.1 (Risk of a learned linear model)**.** *Under the data model above, let $\hat{w}$ be learned from $S \sim \mathcal{D}^m$ by any learner. Then*

$$\begin{aligned}\mathbb{E}_S\left[L_{\mathcal{D}}(h_{\hat{w}})\right] &= \mathbb{E}_S\left[\|\hat{w} - w^*\|_\Sigma^2\right] + \sigma^2 \\ &= \underbrace{\|\mathbb{E}_S[\hat{w}] - w^*\|_\Sigma^2}_{\textit{bias}} + \underbrace{\mathbb{E}_S\left[\|\hat{w} - \mathbb{E}_S[\hat{w}]\|_\Sigma^2\right]}_{\textit{variance}} + \underbrace{\sigma^2}_{\textit{Bayes risk}} \\ &= \|\mathbb{E}_S[\hat{w}] - w^*\|_\Sigma^2 + \mathrm{trace}\left(\mathrm{Cov}_S(\hat{w})\,\Sigma\right) + \sigma^2.\end{aligned}$$

Proof. First equality. Fix S, so that $\hat{w}$ is a constant for the independent test pair. Inserting the model and expanding the square, with $(x^\top a)^2 = a^\top x x^\top a$,

$$\begin{aligned}L_{\mathcal{D}}(h_{\hat{w}}) &= \mathbb{E}_{x,\epsilon}\left[(x^\top \hat{w} - x^\top w^* - \epsilon)^2\right] = \mathbb{E}_{x,\epsilon}\left[(x^\top(\hat{w} - w^*) - \epsilon)^2\right] \\ &= \mathbb{E}_x\left[(\hat{w} - w^*)^\top x x^\top (\hat{w} - w^*)\right] - 2\,\mathbb{E}_{x,\epsilon}\left[\epsilon\, x^\top(\hat{w} - w^*)\right] + \mathbb{E}[\epsilon^2] && \text{(linearity)} \\ &= (\hat{w} - w^*)^\top \mathbb{E}_x[x x^\top](\hat{w} - w^*) - 2\,\mathbb{E}[\epsilon]\,\mathbb{E}_x\left[x^\top(\hat{w} - w^*)\right] + \sigma^2 && (x,\ \epsilon \text{ independent}) \\ &= \|\hat{w} - w^*\|_\Sigma^2 + \sigma^2 && (\mathbb{E}[\epsilon] = 0).\end{aligned}$$

Taking $\mathbb{E}_S$ on both sides gives the claim.

Second equality. Split $\hat{w} - w^* = u + v$ with $u = \hat{w} - \mathbb{E}_S[\hat{w}]$ random and $v = \mathbb{E}_S[\hat{w}] - w^*$ deterministic. Since Σ is symmetric,

$$\|u + v\|_\Sigma^2 = u^\top \Sigma u + v^\top \Sigma v + 2\, u^\top \Sigma v.$$

By linearity $\mathbb{E}_S[u^\top \Sigma v] = \mathbb{E}_S[u]^\top \Sigma v = 0$, because $\mathbb{E}_S[u] = \mathbb{E}_S[\hat{w}] - \mathbb{E}_S[\hat{w}] = 0$. Hence $\mathbb{E}_S\|\hat{w} - w^*\|_\Sigma^2 = \mathbb{E}_S\|u\|_\Sigma^2 + \|v\|_\Sigma^2$, which is variance plus bias.

Third equality. A scalar equals its trace, and the trace is cyclic and commutes with expectation (both are linear):

$$\mathbb{E}_S\left[u^\top \Sigma u\right] = \mathbb{E}_S\left[\,\mathrm{trace}(u^\top \Sigma u)\right] = \mathbb{E}_S\left[\,\mathrm{trace}(u u^\top \Sigma)\right] = \mathrm{trace}\left(\mathbb{E}_S[u u^\top]\,\Sigma\right) = \mathrm{trace}\left(\mathrm{Cov}_S(\hat{w})\,\Sigma\right). \quad \square$$

Bias and variance are ≥ 0 (Σ is positive semidefinite), so no learned linear model beats the Bayes risk σ^2, which h_{w^*} attains. An **unbiased** estimator, $\mathbb{E}_S[\hat{w}] = w^*$, removes the bias term; still, a biased estimator with small enough $\mathrm{Cov}_S(\hat{w})$ can have a smaller expected risk. Balancing the two terms is the **bias-variance tradeoff**.

2.3 Risk of OLS and consistency

From now on $\hat{w}$ is the OLS estimator.

Theorem 2.2 (OLS is unbiased)**.** *If $\hat{\Sigma}$ is invertible with probability* 1*, then $\mathbb{E}_S[\hat{w}] = w^*$.*

Proof. Given $x_1, \ldots, x_m$, the matrix $\hat{\Sigma}$ is fixed, and $\mathbb{E}[y_i \mid x_1, \ldots, x_m] = x_i^\top w^* + \mathbb{E}[\epsilon_i] = x_i^\top w^*$ because ϵ_i is independent of the inputs and has mean 0. Therefore

$$\begin{aligned}\mathbb{E}_S[\hat{w}] &= \mathbb{E}_{x_1,\ldots,x_m}\left[\mathbb{E}\left[\hat{w} \mid x_1, \ldots, x_m\right]\right] && \text{(tower property)} \\ &= \mathbb{E}_{x_1,\ldots,x_m}\left[\hat{\Sigma}^{-1}\left(\frac{1}{m}\sum_{i=1}^m x_i\,\mathbb{E}[y_i \mid x_1, \ldots, x_m]\right)\right] && \text{(linearity)} \\ &= \mathbb{E}_{x_1,\ldots,x_m}\left[\hat{\Sigma}^{-1}\left(\frac{1}{m}\sum_{i=1}^m x_i x_i^\top\right) w^*\right] = \mathbb{E}_{x_1,\ldots,x_m}\left[\hat{\Sigma}^{-1}\hat{\Sigma}\, w^*\right] = w^*. && \square\end{aligned}$$

Theorem 2.3 (Covariance of the OLS estimator). *If $\hat{\Sigma}$ is invertible with probability 1, then*

$$\mathrm{Cov}_S(\hat{w}) = \mathbb{E}_S\left[(\hat{w} - w^*)(\hat{w} - w^*)^\top\right] = \frac{\sigma^2}{m}\,\mathbb{E}_{x_1,\dots,x_m}\left[\hat{\Sigma}^{-1}\right].$$

Proof. The first equality is the definition of Cov_S with $\mathbb{E}_S[\hat{w}] = w^*$ (theorem 2.2). Substituting $y_i = x_i^\top w^* + \epsilon_i$,

$$\frac{1}{m}\sum_{i=1}^m y_i x_i = \Big(\frac{1}{m}\sum_{i=1}^m x_i x_i^\top\Big) w^* + \frac{1}{m}\sum_{i=1}^m x_i \epsilon_i = \hat{\Sigma} w^* + \frac{1}{m}\sum_{i=1}^m x_i \epsilon_i \quad\Longrightarrow\quad \hat{w} - w^* = \hat{\Sigma}^{-1}\Big(\frac{1}{m}\sum_{i=1}^m x_i \epsilon_i\Big).$$

Since $\hat{\Sigma}^{-1}$ is symmetric, $(\hat{w} - w^*)^\top = \big(\frac{1}{m}\sum_j x_j^\top \epsilon_j\big)\hat{\Sigma}^{-1}$, so

$$(\hat{w} - w^*)(\hat{w} - w^*)^\top = \frac{1}{m^2}\,\hat{\Sigma}^{-1}\Big(\sum_{i=1}^m\sum_{j=1}^m \epsilon_i \epsilon_j\, x_i x_j^\top\Big)\hat{\Sigma}^{-1}.$$

The ϵ_i are i.i.d. and independent of the inputs, so $\mathbb{E}[\epsilon_i \epsilon_j \mid x_1, \dots, x_m] = \mathbb{E}[\epsilon_i \epsilon_j]$, which is $\mathbb{E}[\epsilon_i]\mathbb{E}[\epsilon_j] = 0$ for $i \neq j$ and $\mathbb{E}[\epsilon_i^2] = \sigma^2$ for $i = j$. Conditioning on the inputs (tower property) and using linearity,

$$\begin{aligned}
\mathbb{E}_S\left[(\hat{w} - w^*)(\hat{w} - w^*)^\top\right] &= \frac{1}{m^2}\,\mathbb{E}_{x_1,\dots,x_m}\Big[\hat{\Sigma}^{-1}\Big(\sum_{i=1}^m\sum_{j=1}^m \mathbb{E}[\epsilon_i\epsilon_j]\, x_i x_j^\top\Big)\hat{\Sigma}^{-1}\Big] \\
&= \frac{1}{m^2}\,\mathbb{E}_{x_1,\dots,x_m}\Big[\hat{\Sigma}^{-1}\Big(\sigma^2 \sum_{i=1}^m x_i x_i^\top\Big)\hat{\Sigma}^{-1}\Big] \qquad \text{(only } i = j \text{ survives)} \\
&= \frac{1}{m^2}\,\mathbb{E}_{x_1,\dots,x_m}\left[\hat{\Sigma}^{-1}\big(\sigma^2 m \hat{\Sigma}\big)\hat{\Sigma}^{-1}\right] = \frac{\sigma^2}{m}\,\mathbb{E}_{x_1,\dots,x_m}\left[\hat{\Sigma}^{-1}\right].
\end{aligned}$$

□

Gaussian features. Let $\mathcal{D}_\mathcal{X} = \mathcal{N}(0, \Sigma)$ with $\Sigma \in \mathbb{R}^{p\times p}$ invertible (the mean is 0, so $\Sigma = \mathbb{E}[xx^\top]$ as in the data model), and $y = x^\top w^* + \epsilon$ as before. We use without proof the following facts on the Wishart distribution, for $x_1, \dots, x_m$ i.i.d. $\mathcal{N}(0, \Sigma)$. If $m \geq p$, then $\hat{\Sigma}$ is invertible with probability 1; $\hat{\Sigma}$ is Wishart with m degrees of freedom and scale matrix $\frac{1}{m}\Sigma$; $\hat{\Sigma}^{-1}$ is inverse Wishart with m degrees of freedom and scale matrix $m\Sigma^{-1}$; and for $m > p + 1$,

$$\mathbb{E}\left[\hat{\Sigma}^{-1}\right] = \frac{m}{m - p - 1}\,\Sigma^{-1}.$$

Theorem 2.4 (Risk of OLS for Gaussian features and additive noise). *For $\mathcal{D}$ as above and $m > p + 1$,*

$$\mathbb{E}_S\left[L_\mathcal{D}(h_{\hat{w}})\right] = \underbrace{\frac{\sigma^2 p}{m - p - 1}}_{\text{variance}} + \underbrace{\sigma^2}_{\text{Bayes risk}} = \frac{\sigma^2(m-1)}{m - p - 1}.$$

In particular, for fixed p, $\mathbb{E}_S[L_\mathcal{D}(h_{\hat{w}})] \to \sigma^2 = L_\mathcal{D}^*$ as $m \to \infty$.

Proof. Since $m > p + 1 \geq p$, $\hat{\Sigma}$ is invertible with probability 1, so theorems 2.2 and 2.3 apply. Starting from theorem 2.1,

$$\begin{aligned}
\mathbb{E}_S\left[L_\mathcal{D}(h_{\hat{w}})\right] &= \|\mathbb{E}_S[\hat{w}] - w^*\|_\Sigma^2 + \mathrm{trace}\left(\mathrm{Cov}_S(\hat{w})\,\Sigma\right) + \sigma^2 \\
&= 0 + \mathrm{trace}\left(\frac{\sigma^2}{m}\,\mathbb{E}\left[\hat{\Sigma}^{-1}\right]\Sigma\right) + \sigma^2 && \text{(theorems 2.2 and 2.3)} \\
&= \mathrm{trace}\left(\frac{\sigma^2}{m}\cdot\frac{m}{m - p - 1}\,\Sigma^{-1}\Sigma\right) + \sigma^2 && \text{(Wishart mean)} \\
&= \frac{\sigma^2}{m - p - 1}\,\mathrm{trace}(I_p) + \sigma^2 = \frac{\sigma^2 p}{m - p - 1} + \sigma^2 && (\mathrm{trace}(I_p) = p) \\
&= \frac{\sigma^2(p + m - p - 1)}{m - p - 1} = \frac{\sigma^2(m-1)}{m - p - 1}.
\end{aligned}$$

For fixed p, $\frac{m-1}{m-p-1} \to 1$ as $m \to \infty$. □

Consistency. Let $\mathcal{D}$ be a distribution on $\mathcal{X} \times \mathcal{Y}$, ℓ a loss with Bayes risk $L^*_{\mathcal{D}}$, and $\hat{h}_S$ the predictor that a learner $\mathcal{A}$ returns on $S \sim \mathcal{D}^m$.

Definition 2.5 (Consistency)**.** $\mathcal{A}$ is **consistent** for $\mathcal{D}$ and ℓ if $\mathbb{E}_{S\sim\mathcal{D}^m}\left[L_{\mathcal{D}}(\hat{h}_S)\right] \to L^*_{\mathcal{D}}$ as $m \to \infty$. It is **universally consistent** if it is consistent for every distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$.

By theorem 2.4, OLS is consistent for $x \sim \mathcal{N}(0, \Sigma)$, $y = x^\top w^* + \epsilon$ (with p fixed). It is not universally consistent: it fails as soon as the true model is not linear, for instance $y = (x^\top w^*)^2 + \epsilon$.

Remark. Take $x \sim \mathcal{N}(0, \Sigma)$ and $w^* \neq 0$ in this example. Then $\mathbb{E}[y \mid x] = (x^\top w^*)^2$ and $\mathrm{Var}(y \mid x) = \sigma^2$, so $L^*_{\mathcal{D}} = \sigma^2$. For any w put $Z = x^\top w - (x^\top w^*)^2$; the decomposition in the proof of proposition 1.8, $\mathbb{E}[Z^2] \geq (\mathbb{E}Z)^2$, $\mathbb{E}[x] = 0$ and $\mathbb{E}[(x^\top w^*)^2] = w^{*\top}\mathbb{E}[xx^\top]w^*$ give

$$L_{\mathcal{D}}(h_w) = \mathbb{E}[Z^2] + \sigma^2 \geq (\mathbb{E}Z)^2 + \sigma^2 = \left(0 - \|w^*\|^2_\Sigma\right)^2 + \sigma^2 = \|w^*\|^4_\Sigma + \sigma^2.$$

Hence $\mathbb{E}_S[L_{\mathcal{D}}(h_{\hat{w}})] \geq L^*_{\mathcal{D}} + \|w^*\|^4_\Sigma$ for every m, and $\|w^*\|_\Sigma > 0$ since Σ is invertible.

3 Nearest-neighbour classification and consistency

3.1 Nearest-neighbour rules and their asymptotic risk

The rules. Let $\mathcal{X} \subseteq \mathbb{R}^p$ with the Euclidean distance and $S = ((x_i, y_i))_{i=1}^m$. For a test point x, sort the sample points by distance to x and let $\pi_k(x) \in \{1, \dots, m\}$ be the index of the k**-th nearest neighbour** of x, so that

$$\|x - x_{\pi_1(x)}\| \leq \|x - x_{\pi_2(x)}\| \leq \cdots \leq \|x - x_{\pi_m(x)}\|.$$

The **nearest-neighbour (NN) rule** predicts the label of the closest point, $h^{\mathrm{NN}}_S(x) = y_{\pi_1(x)}$; the k**-NN rule** $h^{k\mathrm{NN}}_S(x)$ predicts the majority vote of $y_{\pi_1(x)}, \dots, y_{\pi_k(x)}$, which for labels ± 1 reads $h^{k\mathrm{NN}}_S(x) = \mathrm{sign}\left(\frac{1}{k}\sum_{i=1}^k y_{\pi_i(x)}\right)$. Ties (in distances or in the vote) are not treated. The 1-NN rule is constant on the **Voronoi cells** of the sample (the sets of points closer to x_i than to any other x_j). For finite m the decision boundary depends strongly on S; for large m these rules can represent very complex boundaries (fig. 2).

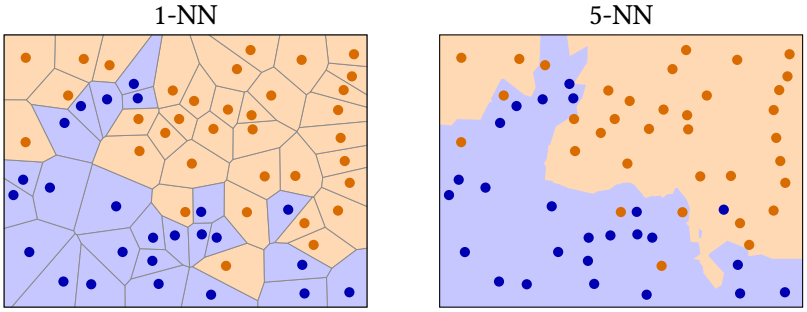


Figure 2: 1-NN and 5-NN decision regions on the same sample.

Example 3.1 (Installation crash, simulated)**.** Take $\mathcal{D}_{\mathcal{X}} = \mathrm{Uniform}[0, 1]$ and $\eta(x) = |1 - 2x|$, whose Bayes risk is $L^*_{\mathcal{D}} = \frac{1}{4}$ (example 1.7). Figure 3 shows the test error of k-NN (5000 test points, average over 25 samples) as m grows. With $k = 1$ or $k = 3$ fixed, the error levels off strictly above $L^*_{\mathcal{D}}$; with $k \approx (\log m)^2$ it approaches $L^*_{\mathcal{D}}$; with $k \approx m/2$ it stays high. The results below explain all three behaviours.

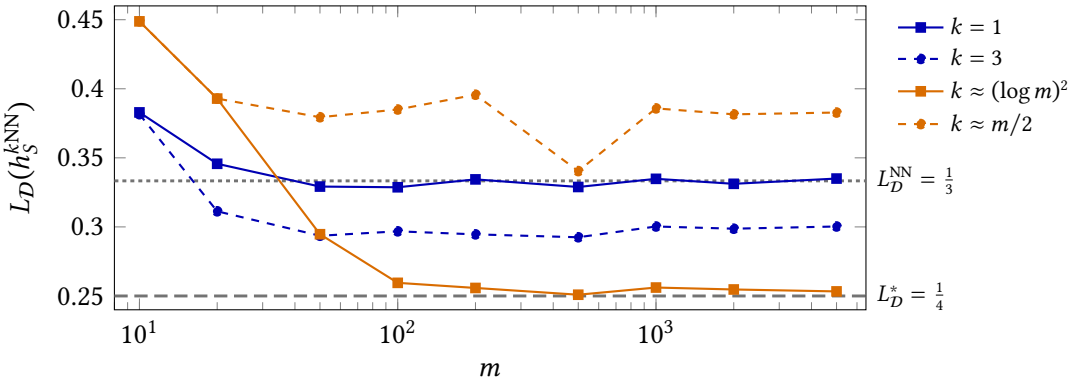


Figure 3: Test error of k-NN on the installation-crash example.

Asymptotic risk. Recall from theorem 1.6 that $L_D^* = \mathbb{E}_x[\min\{\eta(x), 1-\eta(x)\}]$, attained by $h^*(x) = \mathbb{1}\{\eta(x) \geq \frac{1}{2}\}$, which requires knowing η.

Theorem 3.2 (Asymptotic expected risk of the NN rule). *Let* $L_D^{NN} = \mathbb{E}_{x\sim D_X}\big[2\eta(x)(1-\eta(x))\big]$. *Then*

$$\mathbb{E}_{S\sim D^m}\big[L_D(h_S^{NN})\big] \longrightarrow L_D^{NN} \quad \text{as } m \to \infty, \qquad \text{and} \qquad L_D^* \leq L_D^{NN} \leq 2L_D^*.$$

The convergence is proved later in this subsection; the inequalities follow directly.

Proof of the inequalities. Fix x and let $a = \min\{\eta(x), 1-\eta(x)\} \leq \frac{1}{2}$. Since $\{\eta(x), 1-\eta(x)\} = \{a, 1-a\}$, we have $2\eta(x)(1-\eta(x)) = 2a(1-a)$, and $\frac{1}{2} \leq 1-a \leq 1$ gives

$$a = 2a \cdot \tfrac{1}{2} \leq 2a(1-a) \leq 2a.$$

Taking $\mathbb{E}_x$ gives $L_D^* \leq L_D^{NN} \leq 2L_D^*$. □

So with infinitely many data the NN rule is at most twice as bad as the Bayes classifier.

Example 3.3 (Installation crash, NN limit). In example 3.1, $u = |1-2x|$ is again uniform on $[0,1]$: for $t \in [0,1]$, $\mathbb{P}(u \leq t) = \mathbb{P}\big(\frac{1-t}{2} \leq x \leq \frac{1+t}{2}\big) = t$. Hence, with $\mathbb{E}[u] = \frac{1}{2}$ and $\mathbb{E}[u^2] = \frac{1}{3}$,

$$L_D^{NN} = \mathbb{E}\big[2u(1-u)\big] = 2\Big(\frac{1}{2} - \frac{1}{3}\Big) = \frac{1}{3},$$

the level reached by the $k = 1$ curve in fig. 3, and indeed $\frac{1}{4} \leq \frac{1}{3} \leq \frac{1}{2}$.

Theorem 3.4 (Asymptotic expected risk of the k-NN rule). *Let k be fixed. The limit below exists and satisfies*

$$L_D^{kNN} = \lim_{m\to\infty} \mathbb{E}_{S\sim D^m}\big[L_D(h_S^{kNN})\big], \qquad L_D^* \leq L_D^{kNN} \leq \Big(1 + \frac{2}{\sqrt{k}}\Big)L_D^*.$$

The proof (omitted) follows the case $k = 1$ but is more involved. The upper bound approaches L_D^* only as $k \to \infty$, which suggests letting k grow with m.

Why nearest neighbours work. For large m the nearest neighbours of x lie close to x, so (if η is regular) their labels have nearly the same law as the label of x. To make this precise, let $D_X(x;\epsilon) = \mathbb{P}_{x'\sim D_X}(\|x'-x\| \leq \epsilon)$ be the mass of the closed ϵ-ball around x, and let $\mathrm{support}(D_X)$ be the set of $x \in \mathcal{X}$ with $D_X(x;\epsilon) > 0$ for every $\epsilon > 0$. Random vectors z_m **converge in probability** to z if $\lim_{m\to\infty} \mathbb{P}(\|z_m - z\| > \epsilon) = 0$ for every $\epsilon > 0$.

Lemma 3.5 (Nearest neighbours converge in probability). *Let $x \in \mathrm{support}(D_X)$, $x_1, \dots, x_m$ i.i.d. D_X, and $k/m \to 0$ as $m \to \infty$. Then $\mathbb{P}(\|x_{\pi_k(x)} - x\| > \epsilon) \to 0$ for every $\epsilon > 0$. Since $\|x_{\pi_j(x)} - x\| \leq \|x_{\pi_k(x)} - x\|$ for $j \leq k$, all of $x_{\pi_1(x)}, \dots, x_{\pi_k(x)}$ converge to x in probability.*

Proof. Fix $\epsilon > 0$, let $p = \mathcal{D}_{\mathcal{X}}(x;\epsilon) > 0$ and let $Z = \sum_{i=1}^{m} \mathbb{1}\{\|x_i - x\| \le \epsilon\}$ count the sample points in the ϵ-ball. The indicators are independent Bernoulli(p), so $Z \sim \text{Binomial}(m,p)$ and $\mathbb{E}[Z] = mp$. The k-th neighbour is farther than ϵ exactly when fewer than k points lie in the ball:

$$\mathbb{P}\big(\|x_{\pi_k(x)} - x\| > \epsilon\big) = \mathbb{P}(Z \le k-1).$$

For $k = 1$ this is explicit: by independence $\mathbb{P}(Z=0) = \prod_{i=1}^{m} \mathbb{P}(\|x_i - x\| > \epsilon) = (1-p)^m \to 0$. In general we use **Hoeffding's inequality**: for a sum Z of m independent variables with values in $[0,1]$, $\mathbb{P}(Z \le \mathbb{E}[Z] - t) \le e^{-2t^2/m}$ for $t > 0$. Since $k/m \to 0$, for m large enough $\frac{k-1}{m} < \frac{p}{2}$, i.e. $k - 1 < mp - \frac{mp}{2}$, hence

$$\mathbb{P}(Z \le k-1) \le \mathbb{P}\Big(Z \le mp - \frac{mp}{2}\Big) \le \exp\Big(-\frac{2}{m}\Big(\frac{mp}{2}\Big)^2\Big) = e^{-mp^2/2} \xrightarrow[m\to\infty]{} 0. \qquad \square$$

Proof of the convergence in theorem 3.2. We add two assumptions, which are not necessary but simplify the proof:

(A1) $\eta : \mathcal{X} \to [0,1]$ is **uniformly continuous**: for every $\delta > 0$ there is $\epsilon_\delta > 0$ such that $\|x - x'\| \le \epsilon_\delta$ implies $|\eta(x) - \eta(x')| \le \delta$; equivalently, $|\eta(x) - \eta(x')| > \delta$ implies $\|x - x'\| > \epsilon_\delta$.

(A2) $\text{support}(\mathcal{D}_{\mathcal{X}}) = \mathcal{X}$.

Step 1: conditional risk. Under the 0-1 loss, $L_D(h_S^{\text{NN}}) = \mathbb{E}_{(x,y)\sim D}\big[\mathbb{1}\{y \ne y_{\pi_1(x)}\}\big]$. Generate the $m+1$ pairs $(x,y), (x_1,y_1), \dots, (x_m,y_m)$ by first drawing the inputs $x, x_1, \dots, x_m$ i.i.d. from $\mathcal{D}_{\mathcal{X}}$ and then each label independently, equal to 1 with probability η of its input. By the tower property,

$$\mathbb{E}_S\big[L_D(h_S^{\text{NN}})\big] = \mathbb{E}_{S,(x,y)\sim D^{m+1}}\big[\mathbb{1}\{y \ne y_{\pi_1(x)}\}\big] = \mathbb{E}_{x,x_1,\dots,x_m}\Big[\mathbb{E}\big[\mathbb{1}\{y \ne y_{\pi_1(x)}\} \,\big|\, x, x_1, \dots, x_m\big]\Big].$$

Given the inputs, $\pi_1(x)$ is fixed and the labels y, $y_{\pi_1(x)}$ are independent with $\mathbb{P}(y=1) = \eta(x)$, $\mathbb{P}(y_{\pi_1(x)} = 1) = \eta(x_{\pi_1(x)})$. They differ if they are $(1,0)$ or $(0,1)$, so, writing $\eta = \eta(x)$ and $\eta' = \eta(x_{\pi_1(x)})$,

$$\mathbb{E}\big[\mathbb{1}\{y \ne y_{\pi_1(x)}\} \,\big|\, x, x_1, \dots, x_m\big] = \eta(1-\eta') + (1-\eta)\eta'.$$

Step 2: reduction to $|\eta' - \eta|$. Subtracting the integrand of L_D^{NN},

$$\begin{aligned}\eta(1-\eta') + (1-\eta)\eta' - 2\eta(1-\eta) &= \eta - \eta\eta' + \eta' - \eta\eta' - 2\eta + 2\eta^2 \\ &= (\eta' - \eta) - 2\eta(\eta' - \eta) = (1-2\eta)(\eta' - \eta).\end{aligned}$$

Using $|\mathbb{E}z| \le \mathbb{E}|z|$ (Jensen's inequality, $|\cdot|$ convex) and $|1-2\eta| \le 1$ (as $\eta \in [0,1]$),

$$\Big|\mathbb{E}_S\big[L_D(h_S^{\text{NN}})\big] - L_D^{\text{NN}}\Big| = \Big|\mathbb{E}\big[(1-2\eta)(\eta' - \eta)\big]\Big| \le \mathbb{E}\big[|1-2\eta|\,|\eta' - \eta|\big] \le \mathbb{E}_{x,x_1,\dots,x_m}|\eta(x_{\pi_1(x)}) - \eta(x)|.$$

Step 3: splitting at δ. Fix $\delta \in (0,1)$. Split according to whether $|\eta' - \eta| > \delta$; on that event bound $|\eta' - \eta| \le 1$, on its complement by δ:

$$\mathbb{E}|\eta' - \eta| = \mathbb{E}\big[|\eta' - \eta|\,\mathbb{1}\{|\eta' - \eta| > \delta\}\big] + \mathbb{E}\big[|\eta' - \eta|\,\mathbb{1}\{|\eta' - \eta| \le \delta\}\big] \le \mathbb{P}\big(|\eta' - \eta| > \delta\big) + \delta.$$

Step 4: the bad event vanishes. By (A1), $|\eta' - \eta| > \delta$ forces $\|x_{\pi_1(x)} - x\| > \epsilon_\delta$, and the nearest neighbour is farther than ϵ_δ iff all x_i are; conditioning on x and using the case $k=1$ of the proof of lemma 3.5,

$$\mathbb{P}\big(|\eta' - \eta| > \delta\big) \le \mathbb{P}\big(\|x_{\pi_1(x)} - x\| > \epsilon_\delta\big) = \mathbb{E}_x\Big[\mathbb{P}\Big(\min_{i\le m}\|x_i - x\| > \epsilon_\delta \,\Big|\, x\Big)\Big] = \mathbb{E}_x\Big[\big(1 - \mathcal{D}_{\mathcal{X}}(x;\epsilon_\delta)\big)^m\Big].$$

By (A2) every x has $\mathcal{D}_{\mathcal{X}}(x;\epsilon_\delta) > 0$, so the integrand tends to 0 for every x; it lies in $[0,1]$, hence its expectation tends to 0 as well.

Conclusion. Combining Steps 2–4, $\limsup_{m\to\infty} \big|\mathbb{E}_S[L_D(h_S^{\text{NN}})] - L_D^{\text{NN}}\big| \le \delta$ for every $\delta \in (0,1)$, so the limit is 0. $\square$

Without assumptions. The general proof follows Steps 1–2 and then shows $\mathbb{E}|\eta(x_{\pi_1(x)}) - \eta(x)| \to 0$ via part (ii) of the following lemma with $k = 1$ and $f = \eta$, which needs no assumption on η or D_X; theorem 3.4 is proved along the same route. The proof of the lemma is omitted (Devroye, Györfi, Lugosi, Lemmas 5.3–5.4).

Lemma 3.6 (Convergence of a function evaluated at the k nearest neighbours). *Let $f : \mathcal{X} \to \mathbb{R}$ with $\mathbb{E}_{x\sim D_X}|f(x)| < \infty$, and $x, x_1, \dots, x_m$ i.i.d. D_X.*

(i) ***(Stone's lemma)*** $\frac{1}{k}\sum_{i=1}^{k} \mathbb{E}|f(x_{\pi_i(x)})| \le \gamma_p\, \mathbb{E}|f(x)|$, *with* $\gamma_p = \left(1 + \frac{2}{\sqrt{2-\sqrt{3}}}\right)^p - 1$.

(ii) If $k/m \to 0$, then $\mathbb{E}_{x,x_1,\dots,x_m}\left[\frac{1}{k}\sum_{i=1}^{k}|f(x_{\pi_i(x)}) - f(x)|\right] \to 0$ *as* $m \to \infty$.

3.2 Plug-in classifiers and universal consistency

We use consistency and universal consistency as in definition 2.5, for the 0-1 loss.

Plug-in classifiers. The Bayes classifier $h^*(x) = \mathbb{1}\{\eta(x) \ge \frac{1}{2}\}$ (or $\operatorname{sign}(\eta(x) - \frac{1}{2})$ for labels ± 1) is out of reach because η is unknown. A **plug-in classifier** estimates η by some $\hat\eta$ computed from S and predicts

$$\hat h(x) = \mathbb{1}\{\hat\eta(x) \ge \tfrac{1}{2}\} \qquad \big(\text{or } \operatorname{sign}(\hat\eta(x) - \tfrac{1}{2})\big).$$

The idea behind its consistency: if $\hat\eta(x) \to \eta(x)$ for (almost) all x as $m \to \infty$, then $L_D(\hat h) \to L_D^*$; lemma 3.10 makes this quantitative.

Example 3.7 (Naive Bayes). By Bayes' theorem, with $\mathbb{P}(x \mid y = i)$ the class-conditional probability or density,

$$\eta(x) = \frac{\mathbb{P}(y=1)\mathbb{P}(x \mid y=1)}{\mathbb{P}(x)}, \qquad 1 - \eta(x) = \frac{\mathbb{P}(y=0)\mathbb{P}(x \mid y=0)}{\mathbb{P}(x)}.$$

Since $\eta(x) \ge \frac{1}{2} \iff \eta(x) \ge 1 - \eta(x)$, multiplying by $\mathbb{P}(x) > 0$ gives $h^*(x) = \mathbb{1}\{\mathbb{P}(y=1)\mathbb{P}(x \mid y=1) \ge \mathbb{P}(y=0)\mathbb{P}(x \mid y=0)\}$. The plug-in version is $\hat h(x) = \mathbb{1}\{\hat p_1 \hat b_1(x) \ge \hat p_0 \hat b_0(x)\}$, where $\hat p_i$ is the fraction of training points with label i (an easy estimate of $\mathbb{P}(y = i)$) and $\hat b_i(x)$ estimates $\mathbb{P}(x \mid y = i)$. The latter is hard to estimate when x is high-dimensional; **Naive Bayes** assumes that the coordinates of $x = (x^{(1)}, \dots, x^{(p)})$ are independent given the label,

$$\mathbb{P}(x \mid y = i) = \prod_{j=1}^{p} \mathbb{P}\big(x^{(j)} \mid y = i\big),$$

so that $\hat b_i(x)$ only needs one-dimensional estimates.

Weighted-average estimates. For a test point x, let $w_1(x), \dots, w_m(x) \in [0,1]$ be weights with $\sum_{i=1}^m w_i(x) = 1$, computed from x and $x_1, \dots, x_m$ (not from the labels), and set

$$\hat\eta(x) = \sum_{i=1}^{m} \mathbb{1}\{y_i = 1\}\, w_i(x), \qquad \hat h(x) = \mathbb{1}\{\hat\eta(x) \ge \tfrac{1}{2}\}.$$

The NN rule is the case $w_i(x) = \mathbb{1}\{i = \pi_1(x)\}$: then $\hat\eta(x) = \mathbb{1}\{y_{\pi_1(x)} = 1\}$ and $\hat h(x) = y_{\pi_1(x)}$. The k-NN rule is the case $w_i(x) = \frac{1}{k}$ if x_i is one of the k nearest neighbours of x and 0 otherwise:

$$\hat\eta(x) = \frac{1}{k}\sum_{j=1}^{k} \mathbb{1}\{y_{\pi_j(x)} = 1\} = \sum_{i=1}^{m} \mathbb{1}\{y_i = 1\}\, \frac{\mathbb{1}\{i \in \{\pi_1(x), \dots, \pi_k(x)\}\}}{k},$$

and $\hat h(x) = \mathbb{1}\{\hat\eta(x) \ge \frac{1}{2}\}$ is the majority vote.

Theorem 3.8 (Stone's consistency theorem). *The plug-in classifier $\hat{h}$ built from weights w_i as above is universally consistent if, with $x, x_1, \dots, x_m$ i.i.d. $\mathcal{D}_{\mathcal{X}}$:*

(i) there is c such that, for every $f \geq 0$ with $\mathbb{E}_{x\sim D_{\mathcal{X}}}[f(x)] < \infty$, $\mathbb{E}\Big[\sum_{i=1}^{m} w_i(x)\, f(x_i)\Big] \leq c\, \mathbb{E}_{x\sim D_{\mathcal{X}}}[f(x)]$;

(ii) for every $a > 0$, $\lim_{m\to\infty} \mathbb{E}\Big[\sum_{i=1}^{m} w_i(x)\, \mathbb{1}\{\|x_i - x\| > a\}\Big] = 0$;

(iii) $\lim_{m\to\infty} \mathbb{E}\Big[\max_{i\leq m} w_i(x)\Big] = 0$.

The proof is in section 3.3: condition (ii) makes the weights concentrate near x, (iii) forbids any single label from dominating, and (i) is needed only when η is not uniformly continuous.

Universal consistency of k-NN.

Theorem 3.9 (Universal consistency of k-NN). *If $k \to \infty$ and $k/m \to 0$ as $m \to \infty$, then for every distribution $\mathcal{D}$,*

$$\mathbb{E}_{S\sim D^m}\left[L_D(h_S^{k\mathrm{NN}})\right] \to L_D^* \qquad \text{as } m \to \infty.$$

Proof. We check the conditions of theorem 3.8 for the k-NN weights $w_i(x) = \frac{1}{k}\mathbb{1}\{i \in \{\pi_1(x), \dots, \pi_k(x)\}\}$.

(i) The sum only picks the k nearest neighbours, $\sum_{i=1}^{m} w_i(x) f(x_i) = \frac{1}{k}\sum_{j=1}^{k} f(x_{\pi_j(x)})$. Taking expectations, Stone's lemma (lemma 3.6(i), with $|f| = f$) gives the bound with $c = \gamma_p$.

(ii) If the k-th neighbour is within distance a, so are the closer ones; hence the fraction of the k neighbours farther than a is 0 unless $\|x_{\pi_k(x)} - x\| > a$, and it is always ≤ 1:

$$\sum_{i=1}^{m} w_i(x)\mathbb{1}\{\|x_i - x\| > a\} = \frac{1}{k}\sum_{j=1}^{k} \mathbb{1}\{\|x_{\pi_j(x)} - x\| > a\} \leq \mathbb{1}\{\|x_{\pi_k(x)} - x\| > a\}.$$

So the expectation is at most $\mathbb{E}_x\left[\mathbb{P}(\|x_{\pi_k(x)} - x\| > a \mid x)\right]$. Since $k/m \to 0$, the inner probability tends to 0 for every $x \in \mathrm{support}(\mathcal{D}_{\mathcal{X}})$ (lemma 3.5), which carries all the mass of $\mathcal{D}_{\mathcal{X}}$; it is bounded by 1, so the expectation tends to 0.

(iii) $\max_i w_i(x) = \frac{1}{k} \to 0$ because $k \to \infty$. □

This explains fig. 3: $k \approx (\log m)^2$ satisfies both conditions and reaches L_D^*; fixed k violates $k \to \infty$ and stays at $L_D^{k\mathrm{NN}}$ (for $k = 1$, at $L_D^{\mathrm{NN}} = \frac{1}{3}$); $k \approx m/2$ violates $k/m \to 0$.

3.3 Proof of Stone's theorem (not examinable)

This subsection is not part of the exam.

Lemma 3.10 (Risk bound for plug-in classifiers). *Under the 0-1 loss, let $\hat{h}(x) = \mathbb{1}\{\hat{\eta}(x) \geq \frac{1}{2}\}$ for a fixed function $\hat{\eta} : \mathcal{X} \to [0,1]$. Then*

$$L_D(\hat{h}) - L_D^* \leq 2\,\mathbb{E}_{x\sim D_{\mathcal{X}}}|\hat{\eta}(x) - \eta(x)| \leq 2\sqrt{\mathbb{E}_{x\sim D_{\mathcal{X}}}\left[(\hat{\eta}(x) - \eta(x))^2\right]}.$$

Proof. By theorem 1.5, at each x a classifier pays $\eta(x)$ for predicting 0 and $1 - \eta(x)$ for predicting 1. Where $\hat{h}(x) = h^*(x)$ the two classifiers pay the same; where they differ, h^* pays the smaller of the two amounts and $\hat{h}$ the larger, which exceeds it by $|(1 - \eta(x)) - \eta(x)| = 2|\eta(x) - \frac{1}{2}|$. Hence

$$L_D(\hat{h}) - L_D^* = 2\,\mathbb{E}_x\Big[\big|\eta(x) - \tfrac{1}{2}\big|\,\mathbb{1}\{h^*(x) \neq \hat{h}(x)\}\Big].$$

If $h^*(x) \neq \hat{h}(x)$, one of $\eta(x)$, $\hat{\eta}(x)$ is $\geq \frac{1}{2}$ and the other $< \frac{1}{2}$, so $\frac{1}{2}$ lies between them and $|\eta(x) - \frac{1}{2}| \leq |\eta(x) - \hat{\eta}(x)|$. Bounding the indicator by 1 gives the first inequality. The second is Jensen's inequality for the convex $z \mapsto z^2$, $(\mathbb{E}|z|)^2 \leq \mathbb{E}[z^2]$, followed by a square root. □

Reduction. Apply lemma 3.10 for each sample S (with $\hat{\eta}$ computed from S), take $\mathbb{E}_S$, and use Jensen once more over (S, x):

$$\mathbb{E}_S\big[L_D(\hat{h})\big] - L_D^* \le 2\,\mathbb{E}_{S,x}|\hat{\eta}(x) - \eta(x)| \le 2\sqrt{\mathbb{E}_{S,x}\big[(\hat{\eta}(x) - \eta(x))^2\big]}.$$

As the left side is ≥ 0, it suffices to show $\mathbb{E}_{S,x}[(\hat{\eta}(x) - \eta(x))^2] \to 0$. We first assume that η is uniformly continuous (A1), which is not necessary but simplifies the argument. Introduce $\tilde{\eta}(x) = \sum_{i=1}^m \eta(x_i)\, w_i(x)$, the estimate with each label replaced by its conditional mean. By $(a+b)^2 \le 2a^2 + 2b^2$ (which is $(a-b)^2 \ge 0$),

$$(\hat{\eta} - \eta)^2 = \big((\hat{\eta} - \tilde{\eta}) + (\tilde{\eta} - \eta)\big)^2 \le 2(\hat{\eta} - \tilde{\eta})^2 + 2(\tilde{\eta} - \eta)^2,$$

and, using $\sum_i w_i(x) = 1$ to write $\eta(x) = \sum_i \eta(x) w_i(x)$,

$$\hat{\eta}(x) - \tilde{\eta}(x) = \sum_{i=1}^m \big(\mathbb{1}\{y_i = 1\} - \eta(x_i)\big)\, w_i(x), \qquad \tilde{\eta}(x) - \eta(x) = \sum_{i=1}^m \big(\eta(x_i) - \eta(x)\big)\, w_i(x).$$

We show that both squared terms tend to 0 in expectation.

Noise term. Let $\xi_i = \mathbb{1}\{y_i = 1\} - \eta(x_i)$. Given the inputs $x, x_1, \dots, x_m$, the weights are fixed and the ξ_i are independent, with $\mathbb{E}[\xi_i \mid \text{inputs}] = \eta(x_i) - \eta(x_i) = 0$ and $|\xi_i| \le 1$. Hence for $i \ne j$, by the tower property, $\mathbb{E}[\xi_i \xi_j w_i w_j] = \mathbb{E}\big[w_i w_j\, \mathbb{E}[\xi_i \mid \text{inputs}]\, \mathbb{E}[\xi_j \mid \text{inputs}]\big] = 0$. Dropping these cross terms, then using $\xi_i^2 \le 1$, $w_i^2 \le w_i \max_j w_j$ and $\sum_i w_i = 1$,

$$\begin{aligned}
\mathbb{E}_{S,x}\big[(\hat{\eta}(x) - \tilde{\eta}(x))^2\big] &= \mathbb{E}\Big[\sum_{i=1}^m \sum_{j=1}^m \xi_i \xi_j\, w_i(x) w_j(x)\Big] = \mathbb{E}\Big[\sum_{i=1}^m \xi_i^2\, w_i(x)^2\Big] \\
&\le \mathbb{E}\Big[\sum_{i=1}^m w_i(x)^2\Big] \le \mathbb{E}\Big[\max_{i \le m} w_i(x) \sum_{i=1}^m w_i(x)\Big] = \mathbb{E}\Big[\max_{i \le m} w_i(x)\Big] \to 0,
\end{aligned}$$

by condition (iii).

Bias term. Jensen's inequality for the convex $z \mapsto z^2$ with weights $w_i \ge 0$ summing to 1, $\big(\sum_i w_i z_i\big)^2 \le \sum_i w_i z_i^2$, gives

$$\mathbb{E}_{S,x}\big[(\tilde{\eta}(x) - \eta(x))^2\big] = \mathbb{E}\Big[\Big(\sum_{i=1}^m w_i(x)\big(\eta(x_i) - \eta(x)\big)\Big)^2\Big] \le \mathbb{E}\Big[\sum_{i=1}^m w_i(x)\big(\eta(x_i) - \eta(x)\big)^2\Big].$$

Fix $\epsilon > 0$. By (A1) there is $a_\epsilon > 0$ with $\|x_i - x\| \le a_\epsilon \Rightarrow |\eta(x_i) - \eta(x)| \le \epsilon$. Split each term with $1 = \mathbb{1}\{\|x_i - x\| \le a_\epsilon\} + \mathbb{1}\{\|x_i - x\| > a_\epsilon\}$, bounding $(\eta(x_i) - \eta(x))^2$ by ϵ^2 on the first event and by 1 on the second:

$$\begin{aligned}
\mathbb{E}\Big[\sum_{i=1}^m w_i(x)\big(\eta(x_i) - \eta(x)\big)^2\Big] &\le \epsilon^2\, \mathbb{E}\Big[\sum_{i=1}^m w_i(x)\Big] + \mathbb{E}\Big[\sum_{i=1}^m w_i(x)\, \mathbb{1}\{\|x_i - x\| > a_\epsilon\}\Big] \\
&= \epsilon^2 + \mathbb{E}\Big[\sum_{i=1}^m w_i(x)\, \mathbb{1}\{\|x_i - x\| > a_\epsilon\}\Big].
\end{aligned}$$

The last expectation tends to 0 by (ii) with $a = a_\epsilon$. Hence $\limsup_{m \to \infty} \mathbb{E}_{S,x}[(\tilde{\eta}(x) - \eta(x))^2] \le \epsilon^2$ for every $\epsilon > 0$, so the limit is 0. This proves Stone's theorem for uniformly continuous η, without using condition (i).

General η (sketch). Since η is bounded, for every $\epsilon > 0$ there is a uniformly continuous η^* with $\mathbb{E}_x[(\eta(x) - \eta^*(x))^2] < \epsilon$. Let $\tilde{\eta}^*(x) = \sum_i \eta^*(x_i) w_i(x)$ and split

$$\tilde{\eta} - \eta = (\tilde{\eta} - \tilde{\eta}^*) + (\tilde{\eta}^* - \eta^*) + (\eta^* - \eta).$$

The bias argument applied to η^* gives $\mathbb{E}_{S,x}[(\tilde{\eta}^* - \eta^*)^2] \to 0$, and $\mathbb{E}_x[(\eta^* - \eta)^2] < \epsilon$ by construction. For the first term, Jensen as above and condition (i) with $f = (\eta - \eta^*)^2$ give

$$\mathbb{E}_{S,x}\left[(\tilde{\eta}(x) - \tilde{\eta}^*(x))^2\right] \le \mathbb{E}\Big[\sum_{i=1}^{m} w_i(x)\big(\eta(x_i) - \eta^*(x_i)\big)^2\Big] \le c\,\mathbb{E}_x\left[(\eta(x) - \eta^*(x))^2\right] < c\,\epsilon.$$

With $(a+b+d)^2 \le 3(a^2+b^2+d^2)$, $\limsup_m \mathbb{E}_{S,x}[(\tilde{\eta} - \eta)^2] \le 3(c+1)\epsilon$ for every $\epsilon > 0$, so the bias term tends to 0 without assuming (A1).